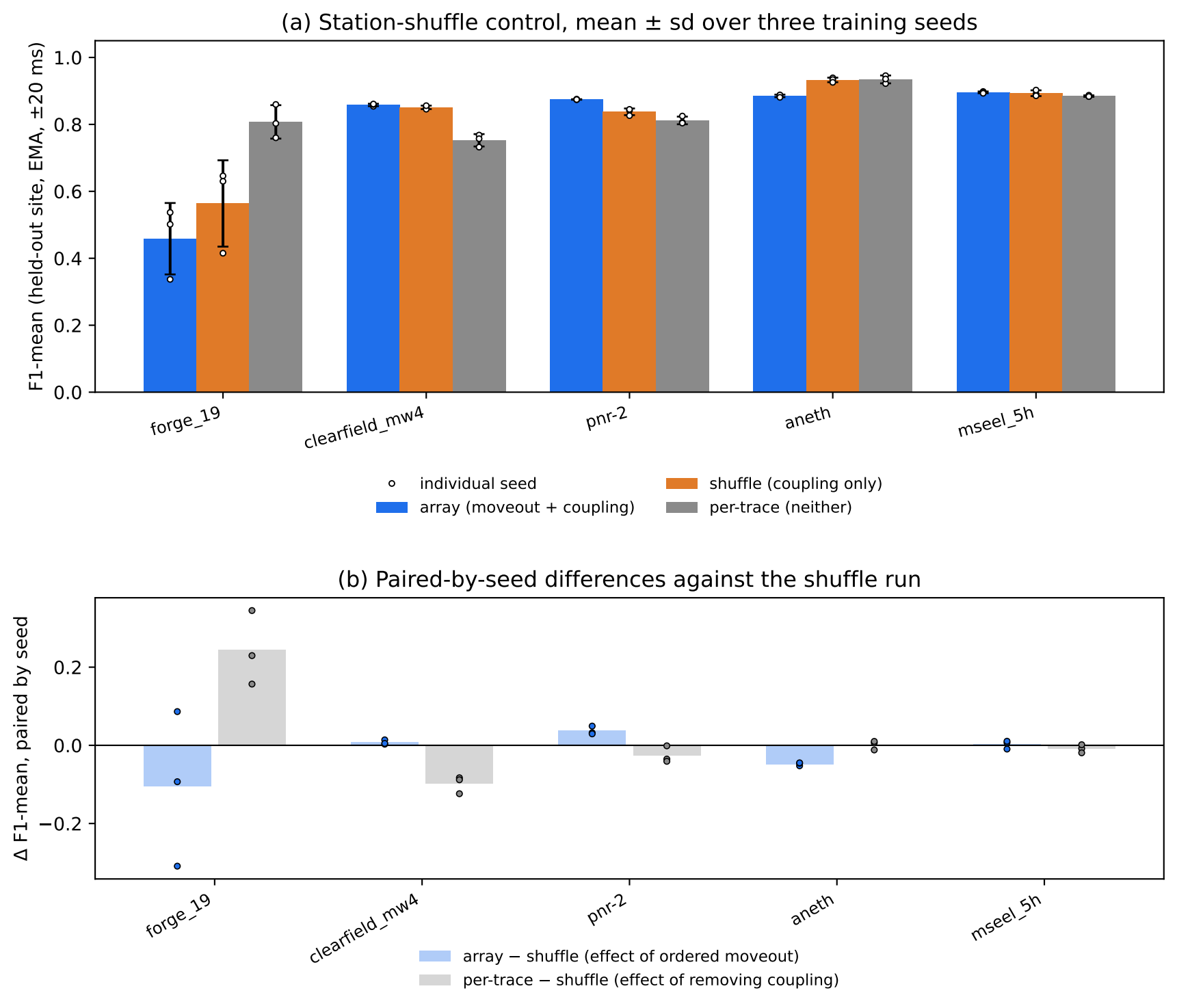
(a) Station-shuffle control, mean ± sd over three training seeds
F1-mean (held-out site, EMA, ±20 ms)
1.0
0.8
0.6
0.4
0.2
0.0
forge_19
clearfield_mw4
pnr-2
aneth
mseel_5h
individual seed
array (moveout + coupling)
shuffle (coupling only)
per-trace (neither)
(b) Paired-by-seed differences against the shuffle run
Δ F1-mean, paired by seed
0.2
0.0
−0.2
forge_19
clearfield_mw4
pnr-2
aneth
mseel_5h
array − shuffle (effect of ordered moveout)
per-trace − shuffle (effect of removing coupling)